VINUK RANAWEERA

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SUMMARY

Data Analyst skilled in Python, SQL, machine learning, and data visualization for predictive modeling and business insights. Built predictive models achieving 87% recall for loan default risk and analyzed 5M+ records to drive strategies projected to increase retention by 10–15%. Experienced in data cleaning, statistical analysis, and visualization using Python, R, and Tableau.

SKILLS

Programming: Python (pandas, numpy, scikit-learn, matplotlib), R (ggplot2, dplyr), SQL
Analysis: Data Cleaning, Exploratory Data Analysis (EDA), Statistical Analysis, Machine Learning, Predictive Modeling
Tools: Tableau, Excel, Google Sheets, BigQuery, MySQL, Jupyter, RStudio, Git, Microsoft Office (Word, PowerPoint)

EXPERIENCE

Web Developer | *Self-Employed/Freelance* **Jan 2016 - Present**
- Designed responsive websites using HTML/CSS, React.js and WordPress, tailored to clients' needs.
- Built and maintained the American Statistical Association NJ Chapter website.

Research Intern | *BCC Geospatial Center of the CUNY CREST Institute* **Jul 2021 - Aug 2022**
- Processed large-scale geospatial imagery datasets using Google Earth Engine APIs, generating GeoTIFF outputs for vegetation monitoring and spatial data visualization.
- Developed Python and Google Earth Engine tutorials for researchers to efficiently analyze satellite imagery.

Teacher Assistant (Junior Video Game Design) **Jan 2017 - Apr 2017**
The Department for Lifelong Learning, Wagner College
- Mentored 25 middle school students in developing 2D video games using Scratch & Unity.

EDUCATION & CERTIFICATIONS

New York University (NYU) Tandon School of Engineering
Master of Science in Computer Engineering **Expected May 2027**

MIT Professional Education
Applied Data Science Program: Leveraging AI for Effective Decision-Making **Aug 2024**
- Applied machine learning to optimize e-commerce product recommendations and analyze marketplace behavior.
- Developed a classification model predicting train passenger satisfaction, ranking Top 5 in model accuracy in the program's final hackathon.

Grove School of Engineering, The City College of New York (CUNY)
Bachelor of Science in Computer Science **Dec 2023**
- *Honors*: Cum Laude, Dean's List
- *Relevant Coursework*: Data Structures, Algorithms, Software Design, Assembly Programming, Discrete Math, Probability Theory, Numerical Analysis in Programming, Modeling Complex Systems, Machine Learning

Coursera
IBM iOS and Android Mobile App Developer Certificate **Aug 2025**
Google Data Analytics Professional Certificate **Jan 2024**
- Acquired skills in data analysis and visualization using Spreadsheets, SQL, R, and Tableau.

Java Programming: Solving Problems with Software by Duke University Certificate **Jul 2019**

PROJECTS

Loan Default Predictions | *Python, Data Cleaning, Predictive Modeling* | Link to Project **Jul 2024 - Aug 2024**
- Developed and evaluated classification models to predict loan default risk, achieving 87% recall on test data.
- Improved performance through feature engineering and hyperparameter tuning to reduce false negatives.

Potential Customer Predictions | *Python, Jupyter, Machine Learning* | Link to Project **Jun 2024 - Jul 2024**
- Cleaned and analyzed lead generation datasets to uncover key factors influencing customer conversion rates.
- Implemented and optimized Decision Tree and Random Forest models to identify high-value prospects.

Cyclistic Bike Share Case Study | *R, RStudio, EDA* | Link to Project **Jan 2024 - May 2024**
- Analyzed over 5 million rider records to identify behavioral trends among casual riders and annual members.
- Delivered data-driven recommendations projected to increase membership subscriptions by 10–15%.